# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 11: **Exploring Some Geometric Themes** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** In the Sierpinski triangle, the number of small shaded triangles after stage 1 is: 1

**(a)** 9 **(b)** 6 **(c)** 3 **(d)** 2

**2.** In the Sierpinski triangle, the number of small shaded triangles after stage 5 is: 1

**(a)** 243 **(b)** 242 **(c)** 486 **(d)** 246

**3.** The number of faces of a cuboid is: 1

**(a)** 6 **(b)** 5 **(c)** 7 **(d)** 8

**4.** The number of faces of a square pyramid is: 1

**(a)** 4 **(b)** 5 **(c)** 7 **(d)** 6

**5.** The number of faces of a pentagonal prism is: 1

**(a)** 8 **(b)** 7 **(c)** 9 **(d)** 6

**6.** A solid has 5 faces and 5 vertices. Using $F + V - E = 2$, the number of edges is: 1

**(a)** 10 **(b)** 8 **(c)** 12 **(d)** 6

**7.** Each side of a Koch curve is replaced at each stage by how many segments? 1

**(a)** 4 **(b)** 5 **(c)** 2 **(d)** 3

**8.** A cylinder has how many curved surfaces? 1

**(a)** 3 **(b)** 0 **(c)** 2 **(d)** 1

**9.** The top view of a cylinder standing on its base is a: 1

**(a)** triangle **(b)** circle **(c)** square **(d)** rectangle

**10.** The number of vertices of a cuboid is: 1

**(a)** 4 **(b)** 8 **(c)** 12 **(d)** 6

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 10–15 words each.*

**11.** What is a fractal? Give an example. 2

**12.** How many triangles are shaded in stage 3 of the Sierpinski triangle? 2

**13.** What is a net of a solid? Draw the net of a cube in words. 2

**14.** State the number of faces, edges and vertices of a cuboid. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 30–40 words each.*

**15.** A Sierpinski triangle starts with one shaded triangle of side 1 unit. After stage 1, find the number of shaded triangles and the side of each. 3

**16.** A Sierpinski triangle starts with one shaded triangle of side 1 unit. After stage 2, find the number of shaded triangles and the side of each. 3

**17.** A Sierpinski triangle starts with one shaded triangle of side 1 unit. After stage 3, find the number of shaded triangles and the side of each. 3

**OR**

A Sierpinski triangle starts with one shaded triangle of side 1 unit. After stage 4, find the number of shaded triangles and the side of each. 3

**18.** A Sierpinski triangle starts with one shaded triangle of side 1 unit. After stage 5, find the number of shaded triangles and the side of each. 3

**OR**

A Sierpinski triangle starts with one shaded triangle of side 1 unit. After stage 6, find the number of shaded triangles and the side of each. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 80–100 words each.*

**19.** (i) Explain how to construct the Sierpinski triangle. 5
(ii) Count the shaded triangles in the first four stages.
(iii) What pattern do you see?

**OR**

(i) Define a polyhedron. 5
(ii) Verify Euler's formula for a cube and a square pyramid.

**20.** **Case study:** A box maker needs a net for a cuboid box 10 cm × 6 cm × 4 cm. 5
(i) List the sizes of the six rectangles.
(ii) Find the total area of cardboard.
(iii) Why will flaps need extra?

**OR**

(i) State the steps to draw the Koch snowflake. 5
(ii) If the original side is 9 cm, find the side length and number of sides after stage 2.

## END OF QUESTION PAPER — Total: 40 Marks

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